

JKBOSE 2026

CLASS 11TH

Plant Growth and Development

LECTURE-04

BOTANY

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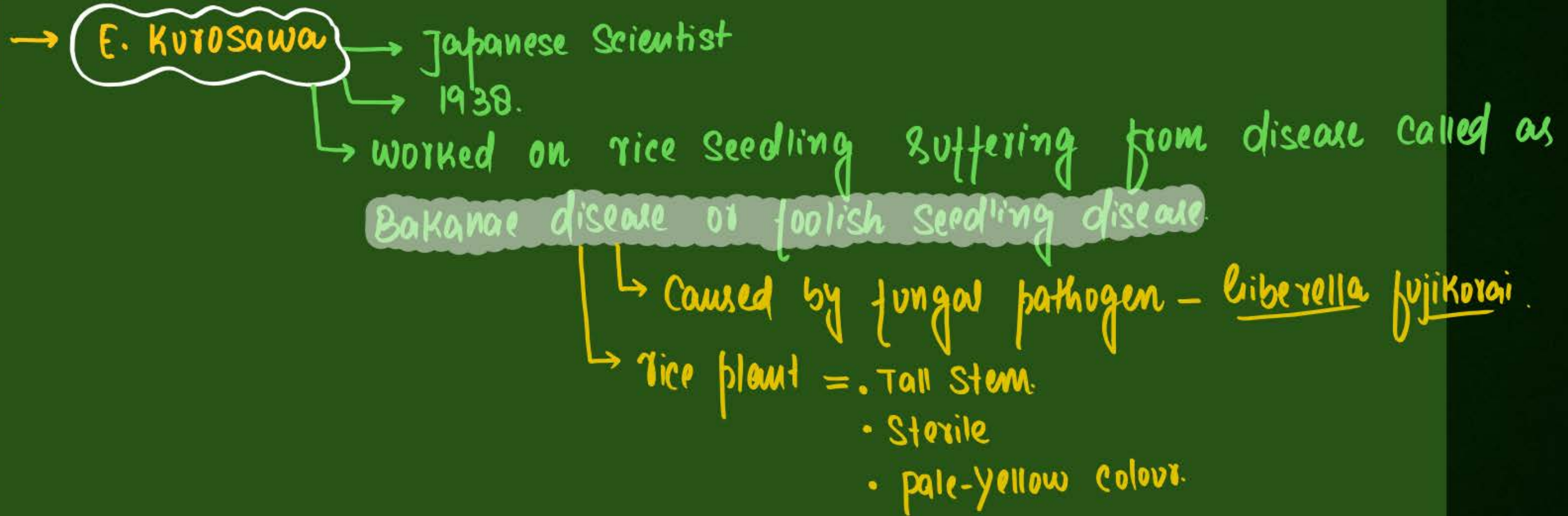
Topics To Covered

- 1** Gibberlins
- 2** Cytokinin

Gibberlins

- Gibberlic acid.
- Acidic in nature.
- Precursors: Terpenes

Discovery



E. Kurosawa → used a sterile filtrate of this fungi.

↓
he applied this sterile filtrate on healthy rice seedling.

↓
Result: Increase in stem length

⊛ later on, it was observed that this internodal elongation of stem is caused by an acidic substance called as gibberellic acid (GA).

NOTE

1. > 100 types of gibberellic acid are discovered namely GA₁, GA₂, GA₃, ...
2. GA₃
 - first discovered gibberellic acid.
 - most studied form.

Role of Gibberlins



- ①  It causes elongation of axis in grapes thus increases grapes production.
- ② It causes elongation in fruit size (ex. Apple, pear). This increases their market value.
- ③ It delays senescence of fruits due to which fruits remain on plants for longer period of time. This increases their market period.
- ④ It causes internodal elongation of sugarcane. (About 20 tonnes per acre increased by spraying GA).
- ⑤ It hastens maturity period in many Conifers which results in early seed production.
- ⑥  It causes bolting (i.e., internode elongation just before flowering) in plants having rosette habit ex. Cabbage, Beet.

⑦. Starch $\longrightarrow$ Maltose $\longrightarrow$ Glucose $\longrightarrow$ Ethanol

It is used in brewing industry to speed-up the malting process.

$\rightarrow$ Here αA_3 is used.

Cytokinin

- Only plant hormone which is basic in nature.
- It is derivative of Adenine (Amino purine) $\rightarrow$ N⁶ furfuryl amino purine.

Discovery

F. Skoog and his workers

- worked on tissue culture.
- observed that internodal stem of tobacco plant forms **CALLUS**



In presence of Auxin only
 $\downarrow$
Callus did not proliferate.

Auxin + Extract of Vascular cambium
+ Yeast Extract + Coconut H₂O
+ DNA
 $\downarrow$
Callus proliferates.

→ **Skoog and Miller (1955)** → identified and crystallised the cytokinesis (cell division)

promoting substance and called it

Kinetin (modified form of Adenine)

→ **Miller** → He first isolated cytokinin (Kinetin) from **Herring** sperm cell DNA.

↓
Fish.

→ plant based cytokin = **zeatin** → isolated from

- Corn Kernel
- Maize
- Coconut H₂O.

(NOTE) Cytokinin is majorly produced in those plant parts which are actively involved in cell division like RAM, SAM, Vascular Cambium etc.

Role of Cytokinin

① It produces → 1. New leaves.

2. Chloroplast in leaves.

3. Adventitious shoots (lateral shoots).

② It increases nutrient metabolism due to it delays Senescence.

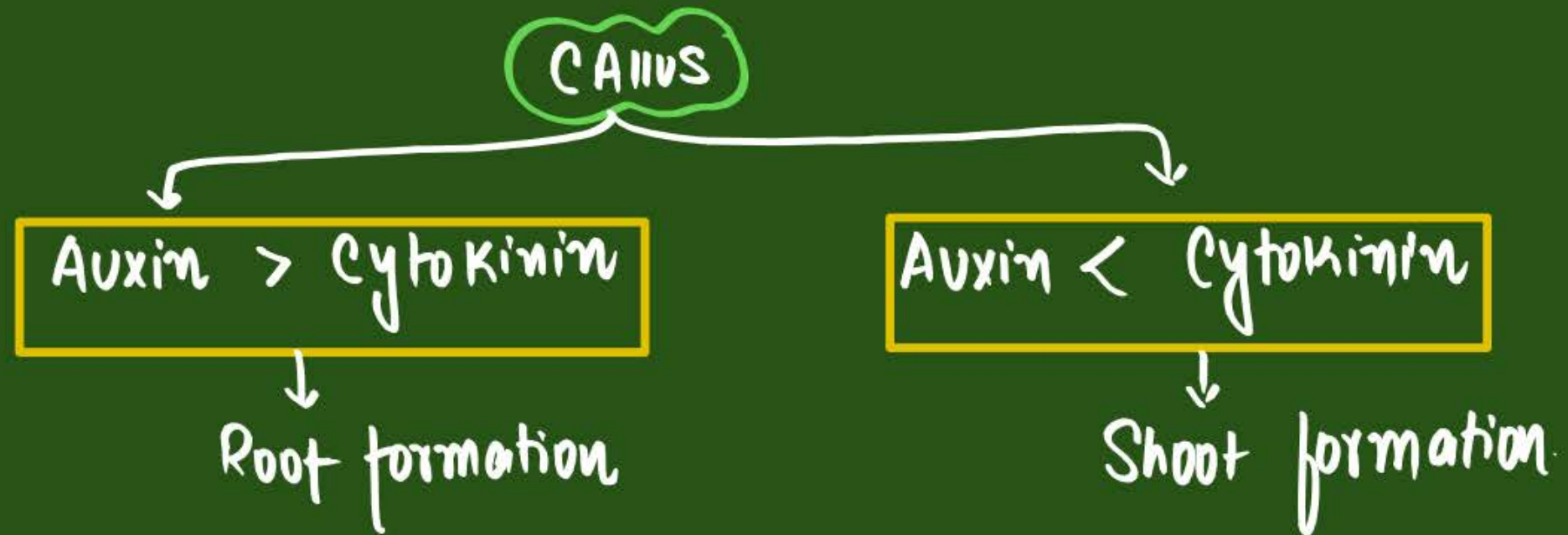
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This is called as Richmond Lang Effect.

• Thus cytokin is AKA anti-ageing hormone

③ It breaks apical dominance by activating the growth of axillary buds.

↓
This role of cytokinin is Antagonistic (opposite) to Auxin.

- ④. It promotes cell division (Synergistic to Auxin).
- ⑤. It increases shelf-life of cut vegetables, cut-fruits etc.
- ⑥. Cytokinin is used in tissue culture.



QUESTIONS

Differentiation of shoot is controlled by

Shoot

Cyt. ↑↑
Auxin ↓↓

$\frac{\text{Cyt.} \checkmark}{\text{Auxin}}$ ↑↑

- 1 High auxin: cytokinin ratio
- ✓ 2 High cytokinin : auxin ratio
- 3 High gibberellin : auxin ratio
- 4 High gibberellin: cytokinin ratio

QUESTIONS

Elongation or lengthening of rosette plants is obtained by treating it with

1 cytokinin

2 gibberellin

3 zeatin

4 auxin

QUESTIONS

Nullification of genetic dwarfism is possible by spraying with

- 1 zeatin
- 2 auxin
- ✓ 3 gibberellin
- 4 cytokinin

HOMework

Revision





Thank
You